

Aashi

Backend Developer · Java · Spring Boot · MySQL
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EDUCATION

B.Tech — Computer Science & Engineering 2023 – 2027

Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow | CGPA: 8.81 / 10

PROFESSIONAL SUMMARY

Backend Developer (B.Tech CSE, AKTU 2027) with hands-on experience building and deploying production-grade systems using Java, Spring Boot, Spring Security, and MySQL. Developed a secure task management backend with JWT Authentication, Role-Based Access Control, cloud database integration (Aiven MySQL), and deployed on Render with environment-variable-based configuration. Built a real-time queue management system with priority scheduling and ACID-compliant transactions, and an advanced banking ledger with row-level locking, deadlock prevention, and idempotency protection. Strong foundation in Data Structures & Algorithms (150+ LeetCode problems), OOP, and DBMS. Seeking a Software Engineering Role to apply and grow these skills in a real-world product environment.

TECHNICAL SKILLS

Languages: Java, C++, C, Python, JavaScript

Backend: Spring Boot, Spring Security, JWT Authentication, REST APIs, Spring Data JPA, Hibernate

Database: MySQL, Aiven Cloud Database — Normalization (3NF), Indexing, Transactions, Row-level Locking, Pagination, Sorting, Filtering

Testing: JUnit 5, MockMvc, Integration Testing, Spring Boot Test

DevOps & Deployment: Git, GitHub, Docker, Render, Environment Variables, Maven

Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS

Tools: Swagger/OpenAPI, Spring Boot Actuator, Postman

PROJECTS

Smart Task Management System [GitHub →](#) [Live Demo →](#)

Java · Spring Boot · Spring Security · JWT · MySQL · Docker · Render · Aiven

- Developed a production-ready Task Management REST API with Spring Boot, Spring Security, JWT Authentication, Hibernate/JPA, and MySQL.
- Implemented Role-Based Access Control (ADMIN/USER) and Ownership-Based Authorization using JWT tokens and Spring Security.
- Designed RESTful APIs for user registration, login, task management, file upload/download, with Pagination, Sorting, and Filtering.
- Integrated Aiven-hosted cloud MySQL database with environment-variable-based configuration for secure credential management.
- Added Spring Boot Actuator endpoints for health monitoring and implemented Spring Cache for performance optimization.
- Developed multipart file upload/download functionality with validation and secure file handling.
- Created unit tests and integration tests using JUnit 5, MockMvc, and Spring Boot Test to validate API behavior.
- Containerized and deployed on Render cloud platform — one of few fresher candidates with a publicly accessible live deployment.
- Documented all APIs using Swagger/OpenAPI and applied global exception handling, logging, and layered architecture principles.

Smart Queue Management System [GitHub →](#)

Java · Spring Boot · Spring Data JPA · MySQL

- Built a RESTful backend for real-time token-based queue scheduling supporting both normal and priority users via FIFO and priority logic.

- Designed a normalized relational schema (3NF) with Spring Data JPA for ORM-based database interactions, reducing boilerplate code.
- Implemented ACID-compliant transaction handling during token serving and queue updates, ensuring zero data inconsistency under concurrent requests.
- Followed layered architecture (Controller → Service → Repository) for clean separation of concerns and extensibility.

Banking Ledger System [GitHub →](#)

MySQL · Stored Procedures · DBMS

- Designed an ACID-compliant banking ledger using MySQL stored procedures with debit-credit entries, balance validation, and an immutable transaction log.
- Implemented row-level locking (SELECT ... FOR UPDATE) with deterministic lock ordering to prevent deadlocks during concurrent fund transfers.
- Built idempotency protection using a unique transaction reference constraint, preventing duplicate transfers under retries or network failures.
- Applied CHECK constraints and SQL-level SIGNAL validation for non-negative balance enforcement and defense-in-depth data integrity.

ACHIEVEMENTS & ACTIVITIES

- Solved 150+ Data Structures & Algorithms problems on LeetCode — consistently practising two-pointer, sliding window, recursion, and graph traversal.
- Successfully deployed a Spring Boot application with JWT security and Aiven cloud MySQL on Render using environment-variable-based config — distinguishing achievement vs. local-only fresher projects.
- Studied advanced database concepts independently — concurrency control, lost update prevention, and log-based ledger design — applied directly in the Banking Ledger project.
- Consistently maintaining CGPA of 8.81/10 across 5 semesters at AKTU while building and shipping personal backend projects.
- Quick learner: independently picked up Spring Boot, Spring Security, JWT, Docker, Swagger/OpenAPI, Spring Boot Actuator, and MySQL stored procedures through documentation and project-based practice.